

Territory Briefing: Torino

Evidence-based preliminary infrastructure intelligence for technical meetings, early design discussion and due diligence. ARCUS supports screening; it does not replace site-specific checks or design verification.

REPORT ID	PROJECT CONTEXT	DOMINANT DRIVER	EVENTS
ARCUS-P01-20260603-torino	Bridge	Hydraulic	42
SPATIAL LEVEL	ANALYSIS TYPE	VERSION	GENERATED BY
Province level	Territory Briefing	ARCUS Path 01 v1.0	ARCUS Professional

Executive Summary

ARCUS identifies 42 documented collapse precedents in the selected province and translates the Step 3 hazard-overlay state into operational next steps for bridge interventions. The dominant territorial signal is **Hydraulic exposure**; the leading historical cause is **Hydraulic**.

Recommended use: province-level preliminary screening before design, due diligence or site-specific investigation.

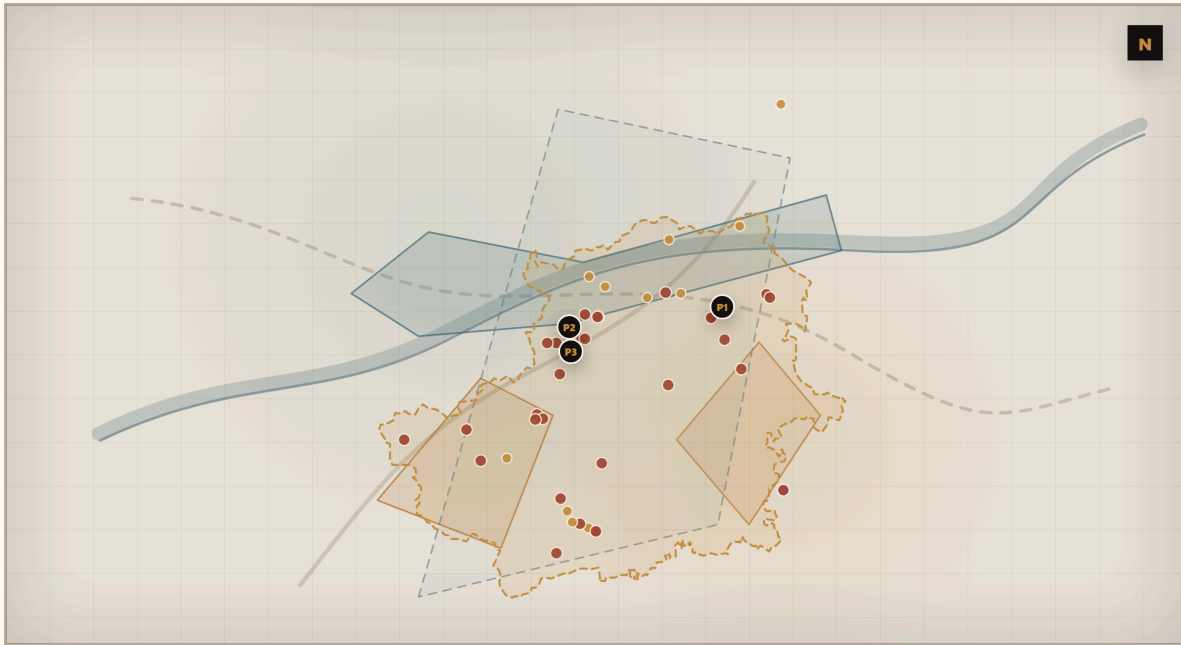
Dominant exposure
Hydraulic exposure / score
97

Historical signal
Hydraulic represents 98% of
the selected evidence.

Priority use
Verify whether the planned
crossing interacts with
active riverbeds or flood-
prone corridors.

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Selected Province & Real Atlas Map



- **TOTAL COLLAPSE** ● **TRIGGERED EVENT** ● **HYDRAULIC OVERLAY**
- **LANDSLIDE OVERLAY** ● **SEISMIC CONTEXT**

Territorial screening map. Not for design scale. P1/P2/P3 markers identify priority events listed in the table below.

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How to Read This Report

ARCUS combines historical collapse records, public hazard context, source-backed evidence and comparable precedents. The output is intended for preliminary screening and planning of follow-up checks, not for structural safety certification.

SELECTED PROJECT CONTEXT

Bridge

Major bridge or bridge intervention: river crossing behaviour, hydraulic compatibility, scour, foundations, piers, abutments, structural concept and inspections.

DOMINANT EXPOSURE

Hydraulic exposure

41 matched ARCUS events. Public overlays are used as screening context and must be followed by site-specific checks.

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Key Indicators

ARCUS EVENTS

42

TOTAL
COLLAPSE

30

TRIGGERED

41

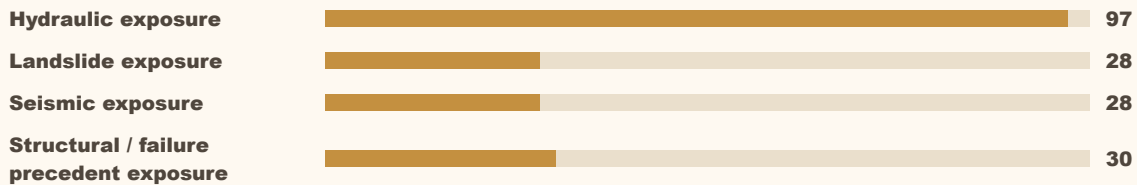
SOURCES

125

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Hazard / Exposure Context

The Step 3 view combines hydraulic WMS, landslide WMS, seismic context and ARCUS historical collapses. The bars below summarise the province-level exposure signal used by the recommendation engine.



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Operational Meaning

PRIMARY FOCUS

Hydraulic compatibility; scour susceptibility; foundation, pier and abutment exposure.

For bridge interventions in the selected province, ARCUS indicates that preliminary attention should focus on hydraulic compatibility and related checks because the dominant indicator is hydraulic exposure. This is a province-level screening signal and must be translated into site-specific checks by qualified professionals.

ATTENTION POINTS

- Hydraulic compatibility
- Scour susceptibility
- Foundation, pier and abutment exposure
- Riverbed dynamics and sediment/debris transport
- Flood/debris scenarios
- Inspection and monitoring accessibility

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Priority ARCUS Precedents

P1/P2/P3 are shown on the Step 3 map. These records are historical analogues for screening and investigation planning, not direct predictions.

P1**B00.10.15 - Castellamonte 2000-10-13 / Hydraulic****TC / High****P2****B00.10.03 - Ala di Stura 2000-10-13 / Hydraulic****TC / High****P3****B00.10.05 - Ala di Stura 2000-10-13 / Hydraulic****TC / High**

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Comparable Precedents

Comparable precedents are selected by severity, trigger, dominant driver, evidence confidence and human impact. For this context, the useful question is not whether a future event is predicted, but which mechanisms and data gaps should be checked before design decisions.

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Interpretation

For bridge interventions in Torino, ARCUS indicates that preliminary investigation should be framed around hydraulic exposure, documented collapse precedents and the operational consequences implied by the selected project context.

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Immediate Screening Actions

ACTION 1

Verify whether the planned crossing interacts with active riverbeds or flood-prone corridors.

ACTION 2

Review hydraulic-triggered total-collapse precedents in the selected province.

ACTION 3

Prioritise hydraulic modelling and scour screening before preliminary design consolidation.

ACTION 4

Check foundation, pier and abutment exposure to riverbed evolution and debris accumulation.

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Data to Request Before Design Decisions

DATA 1

Hydraulic model or preliminary hydraulic study

DATA 2

Riverbed survey / bathymetry where relevant

DATA 3

Historical flood levels

DATA 4

Geotechnical investigation

DATA 5

Foundation concept or available foundation records

DATA 6

Inspection accessibility constraints

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Design-Phase Checks

ACTION 1

Translate province-level screening into site-specific checks by qualified professionals.

ACTION 2

Compare planned intervention assumptions with P1/P2/P3 historical precedents.

ACTION 3

Use public WMS overlays as context, not as detailed hydraulic, geotechnical or seismic modelling.

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Recommended Client Path

1. Verify whether the planned crossing interacts with active riverbeds or flood-prone corridors.
2. Review hydraulic-triggered total-collapse precedents in the selected province.
3. Prioritise hydraulic modelling and scour screening before preliminary design consolidation.
4. Check foundation, pier and abutment exposure to riverbed evolution and debris accumulation.
5. Compare the planned intervention with ARCUS priority precedents around Castellamonte, Ala di Stura, Felleto, Balme, Chialamberto.
6. Request the missing technical data before design or institutional decisions.

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Data Coverage & Limitations

Province-level screening only; not design scale.

ARCUS does not certify structural safety or asset condition.

Public overlays support prioritisation and must be followed by site-specific investigations.



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Methodology Snapshot

ARCUS combines documented collapse events, source reliability, spatial relevance, severity, triggers, cause similarity, territorial hazard context and comparable precedents. The dominant exposure indicator is calculated from Step 3 scores: hydraulic, landslide, seismic and structural/failure precedent exposure.

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Score and Class Legend

P1/P2/P3 identify the top priority historical precedents on the exported Atlas map. Severity TC means total collapse; triggered events indicate an external initiating condition was recorded. Prefer "Failure precedent exposure" unless asset-level structural data are available.

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Short Source Appendix

B24.09.01 / PROTEZIONE CIVILE

Evento meteo-idrologico 05 settembre 2024

B17.06.01 / JOURNAL PAPER

Bridge collapses in Italy across the 21st century: survey and statistical analysis

B08.05.02 / JOURNAL PAPER

Bridge collapses in Italy across the 21st century: survey and statistical analysis

B08.05.02 / REGIONE PIEMONTE

Evento alluvionale 29-30 maggio 2008 – relazione tecnica ufficiale

B05.11.01 / JOURNAL PAPER

Bridge collapses in Italy across the 21st century: survey and statistical analysis

B00.10.03 / ARPA PIEMONTE

Evento alluvionale regionale 13-16 ottobre 2000 – monografia ufficiale

B00.10.03 / JOURNAL PAPER

Bridge collapses in Italy across the 21st century: survey and statistical analysis

B00.10.04 / ARPA PIEMONTE

Evento alluvionale regionale 13-16 ottobre 2000 – monografia ufficiale

The full source table is provided as a separate supporting CSV export.